

# Aeromechanical Model for Robobee Simulator

Much of what is found here can be found in Whitney and Wood’s 2010 paper, ”Aeromechanics of passive rotation in flapping flight”.

The overall simulation framework looks like

- Grab current wing/body state
- Hinge moment balance
- Aerodynamic force and moment calculation.
- Applied Drake forces/torques
- Integration

## 1 Methods

### 1.1 Current Kinematic State

Note that  $\phi$  is the stroke angle (the angle as the wing sweeps back and forth),  $\psi$  is the passive wing pitch angle, and  $\theta$  is the deviation angle (the angle as the wing moves up and down). The total wing angular velocity can be expressed as a vector sum, where

$$\boldsymbol{\omega} = -\dot{\phi}\mathbf{e}_X + \dot{\theta}\mathbf{e}_z + \dot{\psi}\mathbf{e}_x. \quad (1)$$

But since there is no DOF in the  $\theta$  direction, we can set them to 0. Then, the equation simply means it is a sum of the sweep angular velocity and the pitch angle velocity. We then want to break apart  $\boldsymbol{\omega}$  into  $\boldsymbol{\omega}_h$  and  $\boldsymbol{\omega}_x$  (angular velocity around the wing’s own pitch axis). Then, we can define

$$\boldsymbol{\omega}_h = \boldsymbol{\omega} - \boldsymbol{\omega}_x\mathbf{e}_x. \quad (2)$$

(which is literally just saying removing the  $\mathbf{e}_x$  component from rotation). Then, for the no stroke-plane-deviation case,

$$\boldsymbol{\omega}_h = -\dot{\phi}. \quad (3)$$

And after a change of basis (see Appendix 3.1), we can express the angular velocity in the wing-fixed frame as

$$\boldsymbol{\omega}_x = \dot{\psi} - \dot{\phi}\sin\theta \quad (4)$$

$$= \dot{\psi}, \quad (5)$$

again because  $\theta = 0$  in our simplified planar case.

#### 1.1.1 Blade-element kinematics and force application

The Drake/CAD/URDF wing body frame  $B$  is not necessarily aligned with the aerodynamic span/chord axes used by the blade-element model. Let  $\theta_s = \text{span\_axis\_angle\_rad}$  be the fixed angle between the body-frame  $x$ -axis and the aerodynamic span axis. Expressed in the wing body frame,

$$\hat{\mathbf{s}}_B = \cos\theta_s \hat{\mathbf{x}}_B + \sin\theta_s \hat{\mathbf{y}}_B, \quad (6)$$

$$\hat{\mathbf{c}}_B = \sigma_c (-\sin\theta_s \hat{\mathbf{x}}_B + \cos\theta_s \hat{\mathbf{y}}_B), \quad (7)$$

$$\hat{\mathbf{n}}_B = \hat{\mathbf{s}}_B \times \hat{\mathbf{c}}_B, \quad (8)$$

where

$$\sigma_c = \text{sign}(\text{chord\_axis\_sign}).$$

The body-to-aerodynamic-component projection matrix is

$$X_{AB} = \begin{bmatrix} \hat{\mathbf{s}}_B^T \\ \hat{\mathbf{c}}_B^T \\ \hat{\mathbf{n}}_B^T \end{bmatrix}. \quad (9)$$

Thus, for any vector  $\mathbf{v}^B$  expressed in the wing body frame,

$$\mathbf{v}^A = X_{AB} \mathbf{v}^B = \begin{bmatrix} \mathbf{v}^B \cdot \hat{\mathbf{s}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{c}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{n}}_B \end{bmatrix}. \quad (10)$$

The matrix  $X_{AB}$  is fixed for a given wing geometry. The physical pose of the wing is still given by  $R_{WB}$ . Therefore aerodynamic basis vectors are mapped to world coordinates by

$$\hat{\mathbf{s}}_W = R_{WB} \hat{\mathbf{s}}_B, \quad \hat{\mathbf{c}}_W = R_{WB} \hat{\mathbf{c}}_B, \quad \hat{\mathbf{n}}_W = R_{WB} \hat{\mathbf{n}}_B. \quad (11)$$

Equivalently, if

$$R_{BA} = [\hat{\mathbf{s}}_B \quad \hat{\mathbf{c}}_B \quad \hat{\mathbf{n}}_B],$$

then

$$X_{AB} = R_{BA}^T, \quad R_{WA} = R_{WB} X_{AB}^T. \quad (12)$$

For blade element  $i$ , let  $r_i$  be the radial location of the strip and  $\Delta r_i$  its width. The local dynamic pressure and angle of attack are evaluated at a representative blade point  $P_i$ . In this model,  $P_i$  is chosen on the pitch/hinge line:

$$\mathbf{p}_{B_o P_i}^A = \begin{bmatrix} r_i + x_r \\ 0 \\ 0 \end{bmatrix}. \quad (13)$$

Mapping this point into the body and world frames gives

$$\mathbf{p}_{B_o P_i}^B = X_{AB}^T \mathbf{p}_{B_o P_i}^A, \quad (14)$$

$$\mathbf{p}_{B_o P_i}^W = R_{WB} \mathbf{p}_{B_o P_i}^B. \quad (15)$$

The blade-point velocity is then

$$\mathbf{v}_{P_i}^W = \mathbf{v}_{B_o}^W + \boldsymbol{\omega}_h^W \times \mathbf{p}_{B_o P_i}^W. \quad (16)$$

Here  $\boldsymbol{\omega}_h^W$  is the wing angular velocity with the span-axis component removed:

$$\boldsymbol{\omega}_h^W = \boldsymbol{\omega}_B^W - (\boldsymbol{\omega}_B^W \cdot \hat{\mathbf{s}}_W) \hat{\mathbf{s}}_W. \quad (17)$$

The local relative air velocity is first computed in world coordinates and then expressed in the wing body frame:

$$\mathbf{v}_{\text{rel},i}^W = \mathbf{v}_{P_i}^W - \mathbf{v}_{\text{wind}}^W, \quad (18)$$

$$\mathbf{v}_{\text{rel},i}^B = R_{BW} \mathbf{v}_{\text{rel},i}^W. \quad (19)$$

It is then projected into aerodynamic coordinates using the fixed body-to-aero projection matrix:

$$\mathbf{v}_{\text{rel},i}^A = X_{AB} \mathbf{v}_{\text{rel},i}^B = \begin{bmatrix} v_{s,i} \\ v_{c,i} \\ v_{n,i} \end{bmatrix}. \quad (20)$$

The blade-element model uses only the chord-normal part of this relative velocity, so

$$u_i = v_{c,i}, \quad (21)$$

$$w_i = v_{n,i}, \quad (22)$$

$$V_i^2 = u_i^2 + w_i^2, \quad (23)$$

$$\alpha_i = \text{atan2}(w_i, u_i). \quad (24)$$

Using the quasi-steady coefficient model,

$$C_L(\alpha_i) = C_{L,\max} \sin(2\alpha_i), \quad (25)$$

$$C_D(\alpha_i) = \frac{C_{D,\max} + C_{D,0}}{2} - \frac{C_{D,\max} - C_{D,0}}{2} \cos(2\alpha_i), \quad (26)$$

the strip lift and drag magnitudes are

$$\Delta L_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_L(\alpha_i), \quad (27)$$

$$\Delta D_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_D(\alpha_i). \quad (28)$$

In aerodynamic coordinates, the span, chord, and normal unit vectors are simply

$$\hat{\mathbf{e}}_s^A = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_c^A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_n^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The drag direction is opposite the local chord-normal relative velocity:

$$\hat{\mathbf{d}}_i^A = -\frac{u_i \hat{\mathbf{e}}_c^A + w_i \hat{\mathbf{e}}_n^A}{\sqrt{u_i^2 + w_i^2}}. \quad (29)$$

The lift direction is perpendicular to both the span axis and the drag direction:

$$\hat{\mathbf{l}}_i^A = \frac{\hat{\mathbf{e}}_s^A \times \hat{\mathbf{d}}_i^A}{\|\hat{\mathbf{e}}_s^A \times \hat{\mathbf{d}}_i^A\|}. \quad (30)$$

Thus the aerodynamic force on strip  $i$ , expressed in aerodynamic coordinates, is

$$\mathbf{F}_{i,\text{aero}}^A = \Delta L_i \hat{\mathbf{l}}_i^A + \Delta D_i \hat{\mathbf{d}}_i^A. \quad (31)$$

The force is applied at the strip center of pressure  $Q_i$ , not necessarily at the velocity-evaluation point  $P_i$ . The nondimensional distance from the leading edge to the center of pressure is modeled as

$$\hat{d}_{cp,i} = \frac{0.82}{\pi} |\alpha_i| + 0.05. \quad (32)$$

If  $q_{le,i}$  is the chord-axis location of the leading edge relative to the wing root convention, then

$$y_{le,i} = \hat{y}_r \bar{c} + q_{le,i}, \quad (33)$$

$$y_{cp,i} = y_{le,i} - c_i \hat{d}_{cp,i}. \quad (34)$$

Thus the force application point is naturally written in aerodynamic coordinates as

$$\mathbf{p}_{B_o Q_i}^A = \begin{bmatrix} r_i + x_r \\ y_{cp,i} \\ 0 \end{bmatrix}. \quad (35)$$

Mapping this point into the body and world frames gives

$$\mathbf{p}_{B_o Q_i}^B = X_{AB}^T \mathbf{p}_{B_o Q_i}^A, \quad (36)$$

$$\mathbf{p}_{B_o Q_i}^W = R_{WB} \mathbf{p}_{B_o Q_i}^B. \quad (37)$$

The strip force is mapped to body and world coordinates by

$$\mathbf{F}_{i,\text{aero}}^B = X_{AB}^T \mathbf{F}_{i,\text{aero}}^A, \quad (38)$$

$$\mathbf{F}_{i,\text{aero}}^W = R_{WB} \mathbf{F}_{i,\text{aero}}^B. \quad (39)$$

Equivalently,

$$\mathbf{F}_{i,\text{aero}}^W = R_{WB} X_{AB}^T \mathbf{F}_{i,\text{aero}}^A = R_{WA} \mathbf{F}_{i,\text{aero}}^A. \quad (40)$$

Finally, the total aerodynamic force and moment about the wing body origin are

$$\mathbf{F}_{\text{blade}}^W = \sum_i \mathbf{F}_{i,\text{aero}}^W, \quad (41)$$

$$\mathbf{M}_{\text{blade}}^W = \sum_i \mathbf{p}_{B_o P_i}^W \times \mathbf{F}_{i,\text{aero}}^W. \quad (42)$$

The scalar aerodynamic pitch moment about the wing span/pitch axis is obtained by projecting this moment onto  $\hat{\mathbf{s}}_W$ :

$$M_{x,\text{aero}} = \mathbf{M}_{\text{blade}}^W \cdot \hat{\mathbf{s}}_W. \quad (43)$$

### 1.1.2 Blade Velocity

The Drake/CAD/URDF wing body frame  $B$  is not necessarily aligned with the aerodynamic span/chord axes used by the blade-element model. Let  $\theta_s = \text{span\_axis\_angle\_rad}$  be the fixed angle between the body-frame  $x$ -axis and the aerodynamic span axis. Then, expressed in the wing body frame,

$$\hat{\mathbf{s}}_B = \cos \theta_s \hat{\mathbf{x}}_B + \sin \theta_s \hat{\mathbf{y}}_B, \quad (44)$$

$$\hat{\mathbf{c}}_B = \sigma_c (-\sin \theta_s \hat{\mathbf{x}}_B + \cos \theta_s \hat{\mathbf{y}}_B), \quad (45)$$

$$\hat{\mathbf{n}}_B = \hat{\mathbf{s}}_B \times \hat{\mathbf{c}}_B, \quad (46)$$

where  $\sigma_c = \text{sign}(\text{chord\_axis\_sign})$ . The body-to-aerodynamic-component projection matrix is

$$X_{AB} = \begin{bmatrix} \hat{\mathbf{s}}_B^T \\ \hat{\mathbf{c}}_B^T \\ \hat{\mathbf{n}}_B^T \end{bmatrix}. \quad (47)$$

Thus, for any vector  $\mathbf{v}^B$  expressed in the wing body frame,

$$\mathbf{v}^A = X_{AB} \mathbf{v}^B = \begin{bmatrix} \mathbf{v}^B \cdot \hat{\mathbf{s}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{c}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{n}}_B \end{bmatrix}. \quad (48)$$

The world rotation  $R_{WB}$  is still the physical pose of the wing body frame. The span-axis angle only changes the aerodynamic basis vectors inside that body frame, so aerodynamic vectors are mapped to the world frame by first expressing them in body coordinates:

$$\mathbf{v}^W = R_{WB} \mathbf{v}^B. \quad (49)$$

For example,

$$\hat{\mathbf{s}}_W = R_{WB} \hat{\mathbf{s}}_B, \quad \hat{\mathbf{c}}_W = R_{WB} \hat{\mathbf{c}}_B, \quad \hat{\mathbf{n}}_W = R_{WB} \hat{\mathbf{n}}_B. \quad (50)$$

Equivalently, if  $R_{BA}$  has the aerodynamic basis vectors as columns, then

$$R_{BA} = [\hat{\mathbf{s}}_B \quad \hat{\mathbf{c}}_B \quad \hat{\mathbf{n}}_B], \quad (51)$$

$$X_{AB} = R_{BA}^T, \quad (52)$$

$$R_{WA} = R_{WB} X_{AB}^T. \quad (53)$$

For wing strip  $i$ , define the blade point  $P_i$  in the wing body frame as

$$\mathbf{p}_{B_o P_i}^B = (r_i + x_r) \hat{\mathbf{s}}_B + y_{h,i} \bar{\mathbf{c}} \hat{\mathbf{c}}_B. \quad (54)$$

The corresponding world-frame offset and blade-point velocity are

$$\mathbf{p}_{B_o P_i}^W = R_{WB} \mathbf{p}_{B_o P_i}^B, \quad (55)$$

$$\mathbf{v}_{P_i}^W = \mathbf{v}_{B_o}^W + \boldsymbol{\omega}_h^W \times \mathbf{p}_{B_o P_i}^W. \quad (56)$$

As above,  $\boldsymbol{\omega}_h^W$  is the angular velocity with the span-axis component removed:

$$\boldsymbol{\omega}_h^W = \boldsymbol{\omega}_B^W - (\boldsymbol{\omega}_B^W \cdot \hat{\mathbf{s}}_W) \hat{\mathbf{s}}_W. \quad (57)$$

The relative air velocity is then transformed back into the wing body frame:

$$\mathbf{v}_{\text{rel},i}^W = \mathbf{v}_{P_i}^W - \mathbf{v}_{\text{wind}}^W, \quad (58)$$

$$\mathbf{v}_{\text{rel},i}^B = R_{BW} \mathbf{v}_{\text{rel},i}^W. \quad (59)$$

Projecting onto the aerodynamic chord-normal plane gives

$$u_i = \mathbf{v}_{\text{rel},i}^B \cdot \hat{\mathbf{c}}_B, \quad (60)$$

$$w_i = \mathbf{v}_{\text{rel},i}^B \cdot \hat{\mathbf{n}}_B, \quad (61)$$

$$V_i^2 = u_i^2 + w_i^2, \quad (62)$$

$$\alpha_i = \text{atan2}(w_i, u_i). \quad (63)$$

Using the quasi-steady coefficient model,

$$C_L(\alpha_i) = C_{L,\text{max}} \sin(2\alpha_i), \quad (64)$$

$$C_D(\alpha_i) = \frac{C_{D,\text{max}} + C_{D,0}}{2} - \frac{C_{D,\text{max}} - C_{D,0}}{2} \cos(2\alpha_i), \quad (65)$$

so the strip lift and drag magnitudes are

$$dL_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_L(\alpha_i), \quad (66)$$

$$dD_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_D(\alpha_i). \quad (67)$$

The drag direction is opposite the local chord-normal relative velocity, and the lift direction is perpendicular to both the span axis and the drag direction:

$$\hat{\mathbf{d}}_i^B = -\frac{u_i \hat{\mathbf{c}}_B + w_i \hat{\mathbf{n}}_B}{\sqrt{u_i^2 + w_i^2}}, \quad (68)$$

$$\hat{\mathbf{l}}_i^B = \frac{\hat{\mathbf{s}}_B \times \hat{\mathbf{d}}_i^B}{\|\hat{\mathbf{s}}_B \times \hat{\mathbf{d}}_i^B\|}. \quad (69)$$

Therefore, the aerodynamic force on strip  $i$ , expressed in the wing body frame, is

$$\mathbf{F}_{i,\text{aero}}^B = dL_i \hat{\mathbf{l}}_i^B + dD_i \hat{\mathbf{d}}_i^B. \quad (70)$$

The force application point is the center of pressure. First compute

$$\hat{d}_{cp,i} = \frac{0.82}{\pi} |\alpha_i| + 0.05, \quad (71)$$

$$y_{le,i} = \hat{y}_r \bar{c} + q_{le,i}, \quad (72)$$

$$y_{cp,i} = y_{le,i} - c_i \hat{d}_{cp,i}. \quad (73)$$

Then the point where the strip force is applied is

$$\mathbf{p}_{B_o Q_i}^B = (r_i + x_r) \hat{\mathbf{s}}_B + y_{cp,i} \hat{\mathbf{c}}_B. \quad (74)$$

Finally, the strip force is mapped to world coordinates before summing over blade elements:

$$\mathbf{F}_{i,\text{aero}}^W = R_{WB} \mathbf{F}_{i,\text{aero}}^B, \quad (75)$$

$$\mathbf{F}_{\text{blade}}^W = \sum_i \mathbf{F}_{i,\text{aero}}^W. \quad (76)$$

## 1.2 Compute Aerodynamic Coefficients

### 1.2.1 Computing $\alpha$

$$\alpha = f(\psi, \dot{\phi}, \text{stroke direction}, \text{body velocity}, \text{wind}) \quad (77)$$

If  $W$  is the world frame,

$$\mathbf{v}_{rel}^W = \mathbf{v}_{wing}^W - \mathbf{v}_{air}^W \quad (78)$$

$$\mathbf{v}_{rel}^{wing} = R_W^{wing} \mathbf{v}_{rel}^W \quad (79)$$

$$\alpha = \text{atan2}(v_{normal}, v_{chord}) \quad (80)$$

But for wing velocity **through** air, then

$$\alpha = \text{atan2}(v_{normal}, v_{chord}) \quad (81)$$

Another quick note:

$$O' = O + (x_r, y_r). \quad (82)$$

See Fig. 2 for the coordinate axes superimposed on the CAD model. For the latest CAD model, that equals 3.73, -0.86 mm.

### 1.2.2 Computing Aerodynamic Coefficients

We make the quasi-steady assumption, where we pretend that in every instant, the wing is in a steady flow with its current velocity and angle of attack. An empirical fit for the coefficient of lift gives us

$$C_L(\alpha) = C_{L,\max} \sin(2\alpha). \quad (83)$$

For the coefficient of drag, we have

$$C_D(\alpha) = \frac{C_{D,\max} + C_{D,0}}{2} - \frac{C_{D,\max} - C_{D,0}}{2} \cos(2\alpha). \quad (84)$$

For *Drosophila*, we have  $C_{L,\max} = 1.8$ ,  $C_{D,\max} = 3.4$ , and  $C_{D,0} = 0.4$ , from Dickinson et al. (1999).

The normal-force coefficient, which describes the coefficient for force normal to wing surface (and ultimately generates a moment about the pitch hinge), is given by

$$C_N = C_L \cos(\alpha) + C_D \sin(\alpha). \quad (85)$$

## 1.3 Compute blade-element lift and drag

Before starting, note the sign conventions, where  $+\phi$ ,  $+\omega_x$ , and  $+\alpha$  are all positive for the local  $+x$  axis of the wing (i.e. pointing away from the root, along the hinge). For each strip, shown in Fig. 1, we have

$$dF_L = \frac{1}{2} \rho V_i^2 C_L(\alpha) c(r') dr, \quad (86)$$

$$dF_D = \frac{1}{2} \rho V_i^2 C_D(\alpha) c(r') dr, \quad (87)$$

where  $V_i = |\omega_h|(r + x_r)$ . To derive this velocity, observe that a blade element has radial distance  $r + x_r$  (see Fig. 1). Then, the tangential speed (around the stroke,  $X'$  axis), is  $u(r) = |\omega_h|(r + x_r)$ , so  $V_i^2 = \omega_h^2 (r + x_r)^2$ . Furthermore,  $c(r')$  is the chord length at the blade element, where  $r'$  is the distance from the wing root

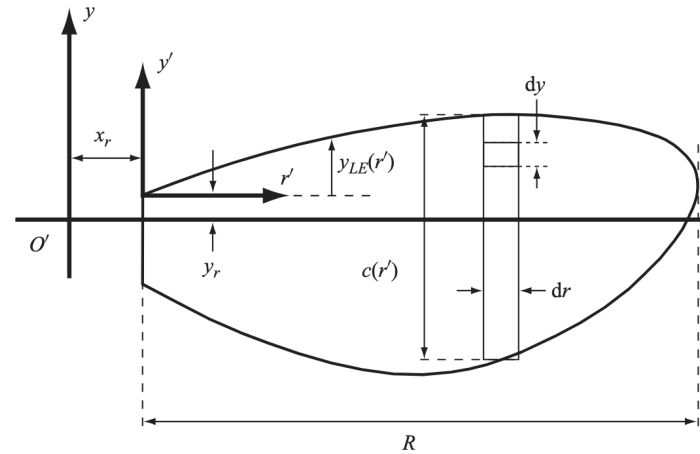


Figure 1: Aeromechanical model of the RoboBee.

to the blade element. Then, the total lift is the integral over the wing span:

$$F_L = \int_0^R dF_L \quad (88)$$

$$= \frac{1}{2} \rho \omega_h^2 C_L(\alpha) \underbrace{\int_0^R (r + x_r)^2 c(r') dr}_{\hat{F}}, \quad (89)$$

where  $\hat{F}$  is the dimensionless wing-shape factor. Full derivation can be found in Appendix 3.2. To non-dimensionalize the integral, set

$$\hat{r} = \frac{r'}{R}, \quad (90)$$

$$\hat{x}_r = \frac{x_r}{R}, \quad (91)$$

$$\hat{c}(\hat{r}) = \frac{c(r')}{\bar{c}}, \quad (92)$$

where  $\bar{c}$  is the mean chord length  $\frac{A}{R}$ . ( $R$  is the wing span).

## 1.4 Compute aerodynamic pitch moment about the hinge

$M_{x,\text{aero}}$  is combination of lift and drag forces, but as a moment. While the integral form given by Whitney and Wood is

$$M_{x,\text{aero}} = \frac{1}{2} \rho \omega_h^2 C_N(\alpha) \int_0^R y_{cp}(r') (r' + x_r)^2 c(r') dr'. \quad (93)$$

Note that  $\alpha$ , the angle of attack. In discrete form, we have

$$M_{x,\text{aero}} \approx \sum_{i=1}^N y_{cp,i} dF_{N,i}. \quad (94)$$

Substituting in Eq. 86, then we have

$$M_{x,\text{aero}} \approx - \sum_{i=1}^N \text{sgn}(\alpha) y_{cp,i} \frac{1}{2} \rho V_i^2 C_N(\alpha_i) c_i \Delta r_i. \quad (95)$$

Note the inclusion of  $\text{sgn}$  because  $\mathbf{F}_N = F_N \mathbf{e}_N = -\text{sgn}(\alpha) F_N \mathbf{e}_z$ .

Here,  $y_{cp}(r')$  is the distance from the leading edge to the center of pressure for the blade element at  $r'$ , and is given by

$$y_{cp} = y_r + y_{LE}(r') - c(r') \hat{d}_{cp}. \quad (96)$$

$y_{LE}$  is basically the distance from the leading edge to the rotation axis, while  $\hat{d}_{cp}$  is the dimensionless distance (think: fraction) from the leading edge to the center of pressure, which is given by Dickson et al. (2006):

$$\hat{d}_{cp} = \frac{0.82}{\pi} |\alpha| + 0.05. \quad (97)$$

## 1.5 Compute passive hinge restoring moment

$M_{x,\text{hinge}} = -\kappa_H \psi$  is a simple linear torsional spring model, where  $k$  is the torsional stiffness. The torsial stiffness is given by

$$\kappa_H = \frac{E_h t_h^3 w_h}{12 L_h}, \quad (98)$$

where  $t_h = 0.0125$  mm,  $w_h = 2.7$  mm, and  $L_h = 0.10$  mm are the thickness, width, and length of the central layer. For Kapton,  $E_h = 2.5$  GPa.

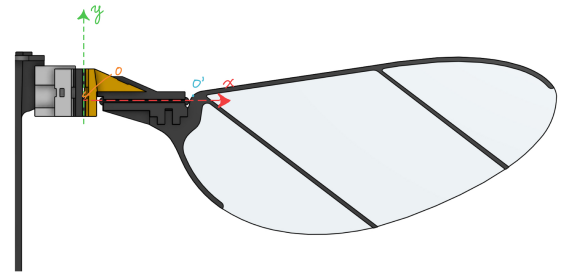


Figure 2: Coordinate axes superimposed on CAD model.

## 1.6 Compute aerodynamic damping moment

$$M_{x,\text{rd}} = -\frac{1}{2}\rho\omega_x|\omega_x|C_{\text{rd}}\bar{c}^4 R\hat{Y}_{\text{rd}}. \quad (99)$$

where

$$\hat{Y}_{\text{rd}} = \int_0^1 \hat{y}_{\text{rd}}(\hat{r}) d\hat{r}. \quad (100)$$

and

$$\hat{y}_{\text{rd}}(\hat{r}) = \frac{1}{4} \left[ |\hat{y}_r + \hat{y}_{LE}| (\hat{y}_r + \hat{y}_{LE})^3 - |\hat{y}_r + \hat{y}_{LE} - \hat{c}| (\hat{y}_r + \hat{y}_{LE} - \hat{c})^3 \right], \quad (101)$$

the non-dimensional location of the effective moment arm of the rotational damping term. Set  $C_{\text{rd}} = 2.0$  as the rotational damping moment coefficient. Again,  $\hat{y}_r$  is the dimensionless distance from the rotation axis to the root of the wing, while  $\hat{y}_{LE}$  is the dimensionless distance from the leading edge (outline of the wing planiform) to the rotation axis.

## 1.7 Compute added mass effect moment

The air inertia resisting wing acceleration, i.e. aerodynamic forces and moments that depend on the body's acceleration, adds another term to the total moment. Whitney and Wood start from the two-dimensional added-mass force and moment on a thin section. After dropping cross-terms that duplicate other quasi-steady blade-element terms, the reduced section equations are

$$Z_0 = -\lambda_z \dot{W}_0 - \lambda_{z\omega} \dot{\omega}_x, \quad (102)$$

$$Y_0 = 0, \quad (103)$$

$$M_0 = -\lambda_{z\omega} \dot{W}_0 - \lambda_\omega \dot{\omega}_x, \quad (104)$$

where  $Z_0$  is the added-mass normal force per unit span,  $M_0$  is the added-mass pitch moment per unit span, and

$$\lambda_z = \pi \rho a^2, \quad (105)$$

$$\lambda_{z\omega} = -\pi \rho a^2 y_h, \quad (106)$$

$$\lambda_\omega = \pi \rho a^2 y_h^2 + \frac{1}{8} \pi \rho a^4. \quad (107)$$

Here  $a = c(r')/2$  is the local semi-chord, and  $y_h$  is the signed offset from the torsional axis to the mid-chord of the local blade-element strip.

Using Whitney and Wood's kinematics, the local normal acceleration of the hinge line is

$$\dot{W}_0(r') = -(r' + x_r) (\dot{\omega}_y - \omega_x \omega_z), \quad (108)$$

where  $(r' + x_r)$  is the distance from the stroke-axis shoulder to the blade element. Radial integration of the reduced section moment gives

$$M_{x,\text{am}} = -\left(\frac{\pi}{4} \rho \bar{c}^3 R^2 \hat{I}_{xy,\text{am}}\right) (\dot{\omega}_y - \omega_x \omega_z) - \left(\frac{\pi}{4} \rho \bar{c}^4 R \hat{I}_{xx,\text{am}}\right) \dot{\omega}_x. \quad (109)$$

Here,  $\hat{I}_{xy,\text{am}}$  and  $\hat{I}_{xx,\text{am}}$  are added-mass geometry integrals analogous to inertial products, and are constants found by

$$\hat{I}_{xy,\text{am}} = \int_0^1 (\hat{r} + \hat{x}_r) \hat{c}(\hat{r})^2 \hat{y}_h(\hat{r}) d\hat{r}, \quad (110)$$

$$\hat{I}_{xx,\text{am}} = \int_0^1 \hat{c}(\hat{r})^2 \left( \hat{y}_h(\hat{r})^2 + \frac{1}{32} \hat{c}(\hat{r})^2 \right) d\hat{r}. \quad (111)$$

### 1.7.1 Applying added mass as a blade-element normal force

For a simulator that applies forces directly to the wing, the translational part of the added-mass effect can be applied strip-by-strip instead of only as a pre-integrated pitch moment. For blade element  $i$ , with radial width  $\Delta r_i$ , local



chord  $c_i$ , semi-chord  $a_i = c_i/2$ , and signed mid-chord offset  $y_{h,i}$ , compute

$$\lambda_{z,i} = \pi \rho a_i^2, \quad (112)$$

$$\lambda_{z\omega,i} = -\pi \rho a_i^2 y_{h,i}, \quad (113)$$

$$\dot{W}_{0,i} = -(r'_i + x_r)(\dot{\omega}_y - \omega_x \omega_z), \quad (114)$$

$$Z_{0,i} = -\lambda_{z,i} \dot{W}_{0,i} - \lambda_{z\omega,i} \dot{\omega}_x. \quad (115)$$

The normal added-mass force on that strip is then

$$\Delta \mathbf{F}_{am,i} = Z_{0,i} \Delta r_i \mathbf{e}_z, \quad (116)$$

with the sign of  $Z_{0,i}$  carrying the direction along the local wing-fixed normal  $\mathbf{e}_z$ . Depending on the simulator's force convention, this may need to be negated when converting from “force of the air on the wing” to “reaction force of the wing on the body.”

The natural application point for this strip force is the local mid-chord location used by the added-mass section model,

$$\mathbf{p}_{am,i} = (r'_i + x_r) \mathbf{e}_x + y_{h,i} \mathbf{e}_y, \quad (117)$$

measured from the shoulder/hinge-frame origin. Applying Eq. 116 at Eq. 117 gives a pitch-axis moment contribution

$$\Delta M_{x,am,i}^{(F)} = y_{h,i} Z_{0,i} \Delta r_i. \quad (118)$$

This captures the moment induced by the translational added-mass normal force. To reproduce the full Whitney–Wood added-mass pitch moment, also include the local pure rotational added-mass couple

$$\Delta M_{x,am,i}^{(\omega)} = -\lambda_{\omega,i} \dot{\omega}_x \Delta r_i, \quad (119)$$

where

$$\lambda_{\omega,i} = \pi \rho a_i^2 y_{h,i}^2 + \frac{1}{8} \pi \rho a_i^4. \quad (120)$$

Thus, if using point forces only, Eq. 116 is the translational/normal added-mass load to apply to the strip. If matching the analytical rotational dynamics exactly, apply both the strip force and the additional strip torque/couple  $\Delta M_{x,am,i}^{(\omega)}$ . Do not also add the pre-integrated  $M_{x,am}$  term as a scalar pitch moment when the strip force is active, or the force-induced moment will be double counted.

## 1.8 Integration

$$I_{xx} \ddot{\psi} = M_x + I_{xy} \ddot{\phi} \cos \psi + \frac{1}{2} I_{xx} \dot{\phi}^2 \sin(2\psi). \quad (121)$$

$$\ddot{\psi} = \frac{M_x + I_{xy} \ddot{\phi} \cos \psi + \frac{1}{2} I_{xx} \dot{\phi}^2 \sin(2\psi)}{I_{xx}}, \quad (122)$$

where  $\ddot{\phi}$  is the stroke angular acceleration, which we can compute from the current stroke angle and velocity/actuator input. See Appendix 4 for derivation of  $I_{xx}$  and  $I_{xy}$ . In particular, assume  $\phi(t)$  (taken at transmission link 1) is commanded from the control algorithms (typically a sin wave with  $\omega = 170\text{Hz}$ , i.e at resonance of transmission mechanism). This should be parametrized in the code. If added mass is applied as blade-element normal forces, the remaining scalar total moment is

$$M_{\text{total}} = M_{\text{aero}} + M_{\text{rot}} + M_{\text{hinge}} \quad (123)$$

But since aero forces originate as translational forces applied to the center of pressure, we can't just apply them purely as moments.

So really,  $M_{\text{aero}}$  needs to be applied as

$$\sum_i \mathbf{r}_{cp,i} \times d\mathbf{F}_i, \quad (124)$$

where

$$\mathbf{r}_{cp,i} = (r + x_r) \mathbf{e}_x + y_{cp}(r) \mathbf{e}_y. \quad (125)$$

## 2 Conclusion and Open Questions

## 3 Appendix

### 3.1 Change of Basis Derivation

The derivation is a change of basis from the mixed-frame angular velocity

$$\boldsymbol{\omega} = -\dot{\phi} \mathbf{e}_{X'} + \dot{\theta} \mathbf{e}_{z'} + \dot{\psi} \mathbf{e}_x \quad (126)$$

to the wing-fixed basis  $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$ . Using the paper's sign convention,

$$\mathbf{e}_{z'} = \sin \psi \mathbf{e}_y + \cos \psi \mathbf{e}_z, \quad (127)$$

$$\mathbf{e}_{X'} = \sin \theta \mathbf{e}_x + \cos \theta (\cos \psi \mathbf{e}_y - \sin \psi \mathbf{e}_z). \quad (128)$$

Substitution gives

$$\begin{aligned} \boldsymbol{\omega} &= -\dot{\phi} [\sin \theta \mathbf{e}_x + \cos \theta (\cos \psi \mathbf{e}_y - \sin \psi \mathbf{e}_z)] \\ &\quad + \dot{\theta} (\sin \psi \mathbf{e}_y + \cos \psi \mathbf{e}_z) + \dot{\psi} \mathbf{e}_x \\ &= (\dot{\psi} - \dot{\phi} \sin \theta) \mathbf{e}_x + (-\dot{\phi} \cos \theta \cos \psi + \dot{\theta} \sin \psi) \mathbf{e}_y \\ &\quad + (\dot{\phi} \cos \theta \sin \psi + \dot{\theta} \cos \psi) \mathbf{e}_z. \end{aligned} \quad (129)$$

Therefore, the wing-fixed angular velocity components are

$$\omega_x = \dot{\psi} - \dot{\phi} \sin \theta, \quad (130)$$

$$\omega_y = -\dot{\phi} \cos \theta \cos \psi + \dot{\theta} \sin \psi, \quad (131)$$

$$\omega_z = \dot{\phi} \cos \theta \sin \psi + \dot{\theta} \cos \psi. \quad (132)$$

The  $-\dot{\phi} \sin \theta$  term in  $\omega_x$  appears because wing deviation projects part of the flapping angular velocity onto the wing pitch axis. In the planar case,  $\theta = 0$ , so  $\omega_x = \dot{\psi}$ .

### 3.2 Non-dimensionalization Derivation

Start with the dimensional spanwise integral from the lift expression:

$$I_F = \int_0^R (r + x_r)^2 c(r') dr. \quad (133)$$

Here  $r$  is the spanwise coordinate used in the blade-element integral, and  $r'$  is the same physical distance measured from the wing root for evaluating the local chord distribution. Thus, for this simplified planar wing model, we take  $r' = r$ .

Using the normalized dimensions for the distance along the wing root, the distance from the wing root to the rotation axis, and the chord length,

$$\hat{r} = \frac{r}{R}, \quad \hat{x}_r = \frac{x_r}{R}, \quad \hat{c}(\hat{r}) = \frac{c(r)}{\bar{c}}, \quad (134)$$

we have

$$r = R\hat{r}, \quad x_r = R\hat{x}_r, \quad c(r) = \bar{c}\hat{c}(\hat{r}), \quad dr = R d\hat{r}. \quad (135)$$

Note that  $(x_r, y_r)$  are the coordinates of the root of the wing (which is basically the hinge). Substituting these into  $I_F$  gives

$$I_F = \int_0^1 (R\hat{r} + R\hat{x}_r)^2 \bar{c}\hat{c}(\hat{r}) R d\hat{r} \quad (136)$$

$$= R^3 \bar{c} \int_0^1 (\hat{r} + \hat{x}_r)^2 \hat{c}(\hat{r}) d\hat{r}. \quad (137)$$

Therefore, the nondimensionalized integral is

$$\hat{F} = \frac{I_F}{R^3 \bar{c}} = \int_0^1 (\hat{r} + \hat{x}_r)^2 \hat{c}(\hat{r}) d\hat{r}. \quad (138)$$

Expanding the square,

$$\hat{F} = \int_0^1 (\hat{r}^2 + 2\hat{x}_r \hat{r} + \hat{x}_r^2) \hat{c}(\hat{r}) d\hat{r} \quad (139)$$

$$= \int_0^1 \hat{r}^2 \hat{c}(\hat{r}) d\hat{r} + 2\hat{x}_r \int_0^1 \hat{r} \hat{c}(\hat{r}) d\hat{r} + \hat{x}_r^2 \int_0^1 \hat{c}(\hat{r}) d\hat{r}. \quad (140)$$

Since  $\bar{c} = A/R$ ,

$$\int_0^1 \hat{c}(\hat{r}) d\hat{r} = \frac{1}{R\bar{c}} \int_0^R c(r) dr = \frac{A}{R\bar{c}} = 1. \quad (141)$$

Define the nondimensional radius moments by

$$\hat{r}_1 = \int_0^1 \hat{r} \hat{c}(\hat{r}) d\hat{r}, \quad (142)$$

$$\hat{r}_2^2 = \int_0^1 \hat{r}^2 \hat{c}(\hat{r}) d\hat{r}. \quad (143)$$

Then the compact form is

$$\hat{F} = \hat{r}_2^2 + 2\hat{x}_r \hat{r}_1 + \hat{x}_r^2. \quad (144)$$

## 4 Moment of Inertia Derivation

The moment of inertia for the left wing  $I_{xx,CM} = 1.3 \times 10^{-6}$  and  $I_{xy,CM} = 2.25 \times 10^{-7}$ , from the CAD model, and around the center of mass. All other elements of the matrix can be approximated as 0 for this pitch equation. However, these need to be expressed around the pitch hinge axis. Assuming the CAD axes are already aligned with the wing-fixed axes used in the derivation, the tensor parallel axis theorem tells us that

$$I_H = I_{CM} + m((\mathbf{d}^T \mathbf{d})\mathbf{1} - \mathbf{d}\mathbf{d}^T), \quad (145)$$

where  $m$  is the wing mass,  $\mathbf{1}$  is the  $3 \times 3$  identity matrix, and  $\mathbf{d}$  is the displacement vector from the wing center of mass to the pitch hinge origin, expressed in the same wing-fixed frame as  $I_{CM}$ . The displacement vector is

$$\mathbf{d} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} = \begin{bmatrix} -7.6171 \\ -0.5949 \\ 0.000 \end{bmatrix}. \quad (146)$$

All distances must be in units consistent with the CAD inertia units. If the inertia is used in SI units, then the displacement vector should be converted from millimeters to meters before applying the theorem.

Expanding the parallel-axis term gives

$$(\mathbf{d}^T \mathbf{d})\mathbf{1} - \mathbf{d}\mathbf{d}^T = \begin{bmatrix} d_y^2 + d_z^2 & -d_x d_y & -d_x d_z \\ -d_y d_x & d_x^2 + d_z^2 & -d_y d_z \\ -d_z d_x & -d_z d_y & d_x^2 + d_y^2 \end{bmatrix}. \quad (147)$$

Therefore, the hinge-frame inertia components needed by the pitch equation are

$$I_{xx,H} = I_{xx,CM} + m(d_y^2 + d_z^2), \quad (148)$$

$$I_{xy,H} = I_{xy,CM} - m d_x d_y. \quad (149)$$

Substituting the displacement vector above,

$$I_{xx,H} = 1.3 \times 10^{-6} + m((-0.5949)^2 + 0^2), \quad (150)$$

$$I_{xy,H} = 2.25 \times 10^{-7} - m(-7.6171)(-0.5949). \quad (151)$$

Equivalently, using the millimeter CAD coordinates as written,

$$I_{xx,H} = 1.3 \times 10^{-6} + 0.35390601 m, \quad (152)$$

$$I_{xy,H} = 2.25 \times 10^{-7} - 4.53141279 m. \quad (153)$$

These are the constants to use for  $I_{xx}$  and  $I_{xy}$  in the pitch dynamics equation:

$$I_{xx} = I_{xx,H}, \quad (154)$$

$$I_{xy} = I_{xy,H}. \quad (155)$$

If the CAD inertia tensor is reported in a frame that is rotated relative to the wing-fixed frame, first rotate it into the wing-fixed frame,

$$I_{CM}^{wing} = R I_{CM}^{CAD} R^T, \quad (156)$$

and then apply the parallel axis theorem above.